NISHCHAY AGRAWAL

@ 03nishchayagarwal@gmail.com in https://www.linkedin.com/in/nishchay-agrawal-157404170/ +919870714012 Khatiyan Street Dhampur, Bijnor, UP, India

EXPERIENCE

Software Development Engineer-2 (Data Engineer-II)

Meesho, India

June 2022 – Present Bangalore, Karnataka,India

Project: Data Platform & Data Intelligence (Data Consumption) Pod

- Build a platform for ingesting and processing multi-terabytes of data daily and also create prototypes and proofs-of-concept for iterative development

Impact And Strength Involved During Project :

- Worked on Zeppelin Databricks SQL connector, when any users run queries on Zeppelin, it will run on Databricks SQL Warehouse. It helps Zeppelin Users to personate queries with RBAC enabled increases the efficiency or decreases the latency of every ANSI SQL query.
- Worked on Springboot API, JPA Service & Developed Custom REST API to generate Personal Access Token for each Zeppelin EKS user so that when any queries will run on Zeppelin EKS using Databricks Spark SQL Interpreter, we can track Query History maintain instrumentation for each User using Service Principal Token.
- Worked on Metabase Cache Warmup Setup which helps to speed up queries by caching only active cards/questions of Metabase instead of caching failed questions of metabase
- Worked on Datahub Governance by using Glossary terms, tags Domains which helps the whole organization to discover or identify which domain or pod/team a particular team belongs to.
- Worked on Data Observability Tasks to identify anomalies present in tables or models using Delta OMS (centralized Delta Log /Transaction Log), Zscore Interquartile Range (IQR) Method.
- Worked on Datahub Spark Lineage for the Spark Custom Jobs Models using Spline Lineage Agent
- Worked on EKS Deployment using Helm Chart for Starburst Presto Zeppelin (Open Source Interactive Notebook) by moving out from ec2-based architecture in order to avoid single point of failure
- Worked on Audit Logging cluster logging management for Databricks using Overwatch Setup
- Worked on the Dashboard Creation of showing metrics on how many Jobs are running or such as cluster costing at notebook level, user level, or command level using Overwatch Audit Logs of Databricks.
- Worked on Datahub for creating Data Lineage for Metabase Dashboards Questions on the Cataloging Tool
- Worked on No-Code-Analytics & Data Modeling in order to provide seamless experience to Business Analysts for Cohort Adhoc Analysis so that they can analyze the metrics dimensions using Superset & instead of writing complex SQL queries
- Worked on OLAP Druid Setup on EKS with Superset for Near Realtime Ingestion of Streaming Data from Kafka-Based Service & Mixpanel Analytics Platform
- Worked on calculating Dimension & Metrics for the Data Models using Logical Spark Plan & SQLGlot and developed the ingestion layer from extracting data from MySQL, MongoDB, events, Change Data Capture (CDC) from Debezium zone & push into silver zone, transform the data & pushing the refined data layer into a gold layer of Hive Metastore (which is linked with S3 Bucket)
- Worked on the calculation of the Number of Instance Pool Used & Billing costs of Databricks Custom Jobs by creating Metabase Dashboard & Delta Table Write in order to capture used Databricks pools during the mega-blockbuster sales period
- Worked on SSO Integration on Datahub, Databricks, Postman Unit testing, Datahub, Model Analytics, Mathematical Statistical Algorithm for Anomalies Findings, deployed docker images on ECR & ECS
- Worked on Lens Kubernetes for Datahub EKS Deployment worked on networking using NodePort so that pods will be reachable from external environment & IP address, Pod Deployment, Replication Controller Service & DNS Mapping using Route 53 which connects user requests to Datahub running on AWS of the production environment

Tech Stack Used :

- PySpark, AWS Databricks, Kubernetes, EKS, Druid OLAP, Superset, Metabase BI Tool, Zeppelin, Datahub Cataloging Tool, Data Governance & Data Observability, Data Modeling (No Code Analytics), Presto, Springboot & JPA Services, Hibernate, Inbuilt Tools, Java, Python, Data Modeling, AWS Tech Stacks, RESTFUL API

Big Data Engineer

Morgan Stanley, India

November 2021 – May 2022 Bangalore, Karnataka,India

Project: Cash Equity & Internal Trade Applications, TEDRA, INSTITUTIONAL SECURITIES GROUP (ISG) Division

- Trade Enrichment Data Reporting Allocations super department in Institutional Securities Technology (IST) Division is responsible for maintaining, distributing, processing, and reporting on trading, revenue, risk, and reference data

Impact And Strength Involved During Project

- Worked on setup the complete infrastructure of Databricks on Azure and Azure DevOps for Code Integration
- Develops ETL Pipeline for fetching data from Snowflake to Databricks using Snowflake Spark connector on Data Lake Platform
- Create the Databricks Runner Script for scheduling or executing the databricks Job,DBFS, clusters operations Snowflake from On-Premise Autosys Job Scheduler by using Databricks Cluster Job REST API 2.1 for Databricks Runner Utility
- Worked on creating the Treadmill on-premise Spark cluster and Domain Objects for front office Data Trade completion data
- Working as Agile Squad Lead for managing the JIRA Kanban Board and DevOps Methodologies
- Setup the Architecture of Databricks Platform on top of Azure using Azure Active Directory Services, Service Principal, Tenant. Managing the Azure Blob Storages Services

Tech Stack Used :

- PySpark, Databricks, SDLC2 Level, Hadoop, Autosys, Snowflake, Data Lake, DB2, Agile, Python Azure, Jenkins Data Pipeline, Bitbucket, Azure, Azure Active Directory, REST API, Inbuilt Spark Based Platform

Big Data Engineer

ZS Associates, India

March 2021 – November 2021 Pune, Maharashtra, India

Project: Amgen Data Strategy Cost Of Sales (COS)

- The Operations Data Strategy – Cost of Sales is an extensive mix of various domains around the manufacturing stages of a product in the supply chain life cycle. These domains delve into the cost of goods manufactured, the cost of goods sold, and eventually cost of sales.

SKILLS

Data Pipeline | Python | Devops | Hadoop | Big Data | Snowflake | Spark | AWS | Zeppelin | NoSQL | Databricks | CI/CD | SQL | Datahub | Kubernetes | Data Warehouse | EKS | Metabase | Springboot | Azure | ETL | Hibernate | Druid | Data Modeling | Data Governance | Data Observability | Superset | Starbust Presto

ADDITONAL SKILLS

Operating Systems | Data Structure | Database | Agile | SDLC | Airflow | Jenkins | Kanban | RESTFUL API

EDUCATION

B.Tech in Computer Science and Engineering

Uttarakhand Technical University

2017 – 2021

Secured Topper Position and Gold Medalist in B.tech Computer Science and Engineering of Uttarakhand Technical University affiliated with more than 100 private & government colleges in Uttarakhand

PERCENTAGE : 91%

XII Board

Central Board of Secondary Education

2016 – 2017

PERCENTAGE : 95%

High School

Central Board of Secondary Education

2014– 2015

PERCENTAGE : 95%

RESEARCH PAPER

- **Nishchay Agrawal and S.Johari, "A Survey on Content-Based Crawling for Deep and Surface Web" in IEEE Conference :** In the Research paper, we have reviewed different web crawlers and classified them on the basis of the information fetched by them. This paper provided a comparative analysis of different web crawlers used for fetching the information based on URL, deep, and surface web.
  IEEE International Conference
  November 2019 Shimla, India
- **Nishchay Agrawal and S.Pant "Web Crawler for Ranking of Websites Based on Web Traffic and Page Views" in Springer International Conference :** In this research paper, we deal with the crawler and ranking of crawled URLs based on web traffic and page views. This approach provides the user with a platform where the user can discover searched content based on popularity.
  Springer International Conference
  September 2020 Dehradun, India

Impact And Strength Involved During Project

- Primarily involved in ODS Cost of Sales project. Worked on creating visualization charts for reporting unit test cases as part of ODS COS
- I was able to leverage all the existing domain collaterals for a smooth onboarding onto the project architecture, domain, and the various logics with minimal hand holding.
- Worked on designing and implementing vacuum Utility to clean up the data files and log files associated with Delta tables using PySpark.
- Created the High-Level Design and Low-Level Design for all the modules of ODS-COS(COGM Actuals, COGM Forecast)
- Created the CI/CD pipeline for deployment automation and data pipeline using AWS and databricks Jobs for ingesting the data coming from source databases, applying transformation logic, and loading the data into the target data warehouse.
- Has aptitude for business understanding and has taken a keen interest in getting acquainted with the client organization and industry data sources.
- I was always eager to learn and expand horizons with various technologies and also had a knack for domain understanding as well.

Tech Stack Used :

- SQL, Databricks, PySpark, Hadoop, Hive, Confluence, Agile, Python, AWS, GitLab, CI/CD, Data Pipeline, R, REVO ETL, Azkaban Scheduler

Big Data Engineer Intern

ZS Associates, India

October 2020 – February 2021 | Pune, Maharashtra, India

Project: Operation Data Strategy Cost Of Sales

- In this project, we deal with actual and forecast data for COGM and COS coming from various data sources such as EDW, and Data Lake (Amgen pharmaceutical client) and deliver optimized solutions using big data technologies, DevOps and AWS.

Impact And Strength Involved During Project

- Primarily involved in ODS Cost of Sales project with Amgen SCM, which is domain heavy in nature.
- Apart from my technical ability, I always ensured there is proper documentation associated with everything worked on which helps any other developer to get easily onboarded.
- Worked on design architecture and code to create a configuration for high-level environment (test and production)for Release-2
- Showing an aptitude for business understanding and has taken a keen interest in getting acquainted with the client organization and industry data sources.

Tech Stack Used:

- Worked on Big Data, Hadoop, Spark, Airflow, Jenkins and Jira, CI/CD, Databricks, Deta Lake, and AWS services.

PROJECTS

Web scraping and ranking of websites based on web traffic

Final year project

November 2020 – Febraury 2021 | Dehradun, India

Impact :

- Based on Php Platform using curl, Alexa ranking, etc.
- It scraps the URLs and hyperlinks and ranked the URLs of the websites based on web traffic and page views using Alexa ranking traffic analysis.

Sentimental Analysis of Twitter for Fetching Top Hashtags of Covid- 19 tweets Using Spark Streaming

University Project

March 2020- April 2020 | Dehradun, India

Impact :

- Based on Apache Spark Streaming, Bahir library, and RDD.
- This Project involves analyzing the tweets of people by going through what they are saying about Covid-19. It provides the top Hashtags related to COVID-19 coronavirus tweets.

STRENGTH

- Quick learner and passionate about working
- Good grasping ability
- Self Motivated
- Confident and Dedicated to my work
- I always show commitment to collective decisions/tasks to be taken in the best interest of the growth

HOBBIES

- Reading books and technical articles
- Dealing and learning with new technology

ACHIEVEMENTS

- Awarded with Certificate of achievement in Spark By IBM
- Certification of Achievement in Spark Developer & Spark Optimization by Databricks
- Certification of Achievement in Big Data by Databricks
- Awarded with Certificate of achievement in Springboot JPA & Spring API
- Certificate of Data warehouse, Data Lake and Data Lakehouse by Databricks
- Best Research Paper Presentation in Springer Conference on "Web Crawler for Ranking of Websites Based on Web Traffic and Page Views"
- Awarded with Gold Medal for being topper in B.tech Computer Science and Engineering of Uttarakhand Technical University by the honorable Governor of Uttarakhand State, Secretary of Technical Education), Minister of Election of Uttarakhand, Vice-Chancellor of UTU, Registrar, Uttarakhand Technical University
- Awarded with Certificate of Snowflake LinkedIn Learning
- Awarded with Certificate of achievement in Kubernetes and EKS
- Awarded with Certificate of Confluent Apache Kafka by Confluent
- Awarded with Certification of Cataloging Tool & Data Governance
- Awarded with E & ICT Certification on Python API Web Services by IIT, Kanpur
- Awarded with Human Resource Development Government of India for securing outstanding performance in CBSE 10th Standard